

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester M. Tech. (Civil-Structural Engineering) Examination

December 2013-January 2014

CL703 Design of Foundation Systems

Date: 04.01.2014, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Describe the Standard Penetration Test used in soil exploration. List the information [04]
that can be obtained by the test when made in (i) Clay, (ii) Sand. Comment on the
correction factors to be used for N-values for;

- (i) Sands for depth below ground level
- (ii) Fine sands under water table.

(b) Write the brief critical note on Vane Shear Test. [03]

Q - 2 Attempt Any One; [14]

- (I) Briefly describe the cyclic plate load test and block vibration test. Also clearly
mention how the dynamic soil parameters can be calculated from these tests.
- (II) Calculate the gross and safe bearing pressure of sand having $\phi = 40^\circ$ and effective
unit weight = 16 kN/m^3 under the following cases:
 - (i) 1.5 m wide strip footing
 - (ii) 1.5m X 1.5m square footing.

The depths of footings are 1.5 m from the GL and water table is 0.5 m below the
footing's base. Assume $F=3$. Take $N_q = 64.2$ and $N_\gamma = 109.4$.

Q - 3 A Plate Load Test (PLT) was conducted on a uniform sand deposit and the data are; [14]

Pressure (kPa)	50	100	200	300	400	500	600
Settlement (mm)	1.5	2	4	7.5	12.5	20	40

The size of the plate was 750mm X 750mm and that of the pit was 3.75m X 3.75m X
1.5m. (i) Find the ultimate bearing capacity from pressure-settlement curve. (ii) A
square footing of 2m X 2m is to be placed at 1.5m depth in this soil and find the
allowable bearing pressure for permissible settlement of 40mm.

OR

Q - 3 (a) A block vibration test yields following data;

[10]

f (rpm)	500	600	700	750	850	950	1000	1200
Az (mm)	0.2	0.6	1.5	2.5	3.2	2.5	1.5	0.6

Plot a graph between operating frequency (f) on X-axis and vertical displacement (Az) on Y-axis and find the following;

- (i) Resonant frequency (fr)
- (ii) Damping factor (D)
- (iii) The coefficient of elastic uniform compression (Cu).

The size of the concrete block is 1.5m X 0.75m X 0.70m. Take the unit weight of concrete as 24 kN/m^3 .

- (b) A heavy wall has total weight of 10,000kN and is embedded in sandy deposit having $\mu = 0.5$. If the magnitude of resultant passive reaction on left face, $P_1 = 281 \text{ kN/m}$, find the vertical reaction (R) at the base. [04]

SECTION - II

- Q - 4 (a) Explain with Sketches the installation stages of Bored pile using bentonite slurry technique along with details of reinforcement & tremie concreting [04]

- (b) Explain the sand compaction pile and stone column method to be adopted for constructing structures on collapsible landfills. [03]

- Q - 5 (a) A SPT was performed at a depth of 20m in a dense sand deposit with a unit weight of 18 kN/m^3 . If the observed N is 48, what is the N-value corrected for overburden? [10]

- (b) Explain the collapse mechanism. State the relation of collapse potential & air voids. Give critical degree of saturation for various types of soils. [04]

OR

- Q - 5 (a) In transmission tower foundation leg is provided with under-reamed pile to take care of uplift forces during wire cut condition. Estimate uplift force. Shaft diameter = 0.25m. Bulb diameter = 3 times shaft diameter, Depth of active zone below GL = 2.5m having $C = 200 \text{ kN/m}^2$, the bulb is made at 3.5m below GL in non-active zone having $C = 150 \text{ kN/m}^2$. Take $\alpha = 0.7$. [10]

- (b) State the factors affecting Group action of pile. Explain converse Feld's rule for estimating pile group efficiency. [04]

- Q - 6 (a) Design the pile cap for single pile from following data: Column load = 50t; column size = (300x400)mm; pile diameter = 300mm; Pile cap size = (600x600x950)mm. [08]

- (b) Explain the pile load test in detail with neat sketch showing all the components. [06]

OR

- Q - 6 (a) A (40x40) cm RCC pile was driven loose & soft deposit 20 m thick in to a stratum of dense sand for a distance of 1m. Submerged density = 0.72 t/m^3 , angle of internal friction = 34° , unconfined strength = 0.6 kg/cm^2 . Estimate frictional resistance & calculate safe bearing capacity (SBC) with FOS = 3 taking base resistance = 97.6 t. What will be the SBC of Group of piles arranged in 4 rows with 4 piles in each row spaced 1.2 m c/c. [08]

- (b) Stating identification & damage criteria of Expansive clay, explain with sketches typical damages that a structure will suffer. Support your answer with at least two case studies. [06]
